PHIL 450: Midterm

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March 9, 2023

1. In this question you will define an operation *sort* that takes a sequence of numbers and sorts them in ascending order, yielding another sequence of numbers. You may assume that the relation $m \leq n$ as already been defined, that is, you may assume that you have defined whether one number less than or equal to another.

   (a) State the two rules for creating sequences of numbers. (3 points)

   (b) Consider the operation, *insert* that takes a natural number, m, and a sequence of natural numbers, t, and returns a sequence, $insert(m, t)$, which is the result of inserting m into the sequence t after the last number in t that m is at least as big as. So $insert(4, (1, 3, 2, 8, 1, 5)) = (1, 3, 2, \mathbf{4}, 8, 1, 5)$: the 4 has been inserted after the $1, 3$ and 2, because it is at least as big as 1, 3 and 2, and it has not been inserted any later, because 8 is bigger than it.

   Using the informal description above, calculate $insert(3, (2, 3, 4))$ and $insert(5, (4, 1, 2, 3, 1, 11))$ (3 points)

   (c) Define $insert(m, t)$ by recursion. Hint: you should hold m fixed, and define it by recursion on t. For the "step" case of the definition, you will need to say what the function outputs in two different cases. (5 points).

   (d) Define by recursion an operation, *sort* that maps a sequence of numbers to another sequence of numbers, that sorts a sequence of numbers in ascending order. So, for instance, $sort((4, 1, 9, 8, 3, 4, 3, 1, 4)) = (1, 1, 3, 3, 4, 4, 4, 8, 9)$. You may assume that your definition of *insert* behaves as it ought to. (Hint: you will only need to use your definition of *insert* above.) (10 points).

2. (a) Show that $\models (A \wedge B) \rightarrow B$. (4 points).

   (b) Show that $A \rightarrow B, B \rightarrow C \models A \rightarrow C$. (4 points)

   (c) Suppose that $A \models C$ and $B \models C$. Show that $A \vee B \models C$. (6 points).

   (d) Show that $Q, P \rightarrow Q \not\models P$ by finding a valuation that makes the premises true but not the conclusion. (5 points).

3. In this question you will need the definition of multiplication and addition. In infix notation they are given as follows. Addition: $k + 0 = k$, $k + sn = s(k + n)$. Multiplication: $k \times 0 = 0$, $k \times sn = (k \times n) + k$. (You may use infix or prefix notation to represent it, as you prefer.)

   (a) Using recursion, define the exponentiation function $exp : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ where $exp(k, n) = k^n = k \times k \times .... \times k$, the result of multiplying k with itself n times. (Hint: fix k and do the recursion on n). (5 points).

   (b) Consider the function f that maps the number n to the result of powering 2 to itself n times. We will simply stipulate that $f(0) = 1$. That is, $f(0) = 1$, $f(1) = 2$, $f(2) = 2^2 (= 4)$, $f(3) = 2^{2^2} (= 2^4 = 16)$, $f(4) = 2^{2^{2^2}} = 2^{16}$ and so on. Now define it using recursion. (10 points).

   (c) Prove by induction that $exp(k, m + n) = exp(k, m) \times exp(k, n)$. You may assume that for any numbers m, n and$k$, $(m \times n) \times k = m \times (n \times k)$. You may also assume that for any numbers m and n, $m \times n = n \times m$.[1] (Hint: hold k and m fixed, and do the induction on n.) (15 points).

[1] You may not need this second assumption if depending on how you defined exponentiation in the first part.